program_top_LR_features

| Program | Interaction | Weight |
|---|---|---|
| 0 | MIF → TNFRSF14 | 0.83381601961147 |
| 0 | HLA-DRB5 → CD4 | 0.801298265009224 |
| 0 | SELPLG → ITGB2 | 0.765445784277238 |
| 0 | LTB → TNFRSF1A | 0.748398010490799 |
| 0 | HLA-F → LILRB1 | 0.743895281131986 |
| 0 | TNFSF13 → TNFRSF1A | 0.709545696759077 |
| 0 | SELPLG → SELL | 0.695465461159389 |
| 0 | FADD → FAS | 0.692403520652979 |
| 0 | TNFSF13 → TNFRSF14 | 0.687428878348453 |
| 0 | APP → LRP10 | 0.686117051281589 |
| 0 | HLA-G → LILRB1 | 0.679177688506596 |
| 0 | HLA-C → CD8B | 0.665340101864442 |
| 0 | HLA-DRB1 → LAG3 | 0.661187560169463 |
| 0 | APP → CD74 | 0.65700504474139 |
| 0 | HLA-A → CD8B | 0.656712902824051 |
| 0 | HLA-B → CD8B | 0.650611627421128 |
| 0 | HLA-DPB1 → LAG3 | 0.649946747199019 |
| 0 | HLA-E → CD8B | 0.648091792060087 |
| 0 | HLA-DPA1 → LAG3 | 0.646675232607353 |
| 0 | SERPINA1 → LRP1 | 0.645696412502104 |
| 1 | VEGFA → NRP1 | 0.715766829651752 |
| 1 | CD99 → CD81 | 0.699454424198847 |
| 1 | TXLNA → STX4 | 0.692531291977251 |
| 1 | TGFB1 → CXCR4 | 0.67946273826266 |
| 1 | CALM1 → PTPRA | 0.677424807067571 |
| 1 | PTGS2 → CAV1 | 0.675267098013152 |
| 1 | ICAM4 → ITGAV | 0.673443130233014 |
| 1 | TIMP2 → CD44 | 0.673143554962808 |
| 1 | VEGFB → ADRB2 | 0.66622271933889 |
| 1 | HBEGF → CD44 | 0.665053574340731 |
| 1 | ICAM4 → ITGB1 | 0.661909639602786 |
| 1 | VEGFA → NRP2 | 0.659908387583547 |
| 1 | TNFSF12 → TNFRSF12A | 0.656006410019695 |
| 1 | VEGFA → SIRPA | 0.655944229555253 |
| 1 | GNAI2 → CAV1 | 0.650798839259977 |
| 1 | TNFSF9 → HLA-DPA1 | 0.647704738737529 |
| 1 | ACTR2 → ADRB2 | 0.644524401366065 |
| 1 | VIM → CD44 | 0.644387469976739 |
| 1 | CALR → ITGAV | 0.643399506758528 |
| 1 | BMP2 → ENG | 0.642594917374504 |
| 2 | ADCYAP1 → RAMP2 | 1.01340182018977 |
| 2 | IAPP → RAMP2 | 1.00614180742894 |
| 2 | GNAI2 → IGF1R | 0.984606756162454 |
| 2 | AVP → GPR20 | 0.964727791469604 |
| 2 | PTHLH → RAMP2 | 0.964165409615704 |
| 2 | GNAI2 → UNC5B | 0.960324775664607 |
| 2 | CNTF → IL6R | 0.946268512274143 |
| 2 | OMG → TNFRSF1B | 0.927937874968863 |
| 2 | AVP → RAMP2 | 0.914552203281865 |
| 2 | HLA-B → KLRD1 | 0.904176218991694 |
| 2 | HLA-E → KLRK1 | 0.898651862131838 |
| 2 | CEACAM5 → CD8A | 0.891748669163855 |

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|---|---|
| 2 CCL11→CCR5 | 0.888743417597135 |
| 2 ICAM4→ITGB3 | 0.878999681233468 |
| 2 PRSS3→F2R | 0.876903532449941 |
| 2 OMG→LINGO1 | 0.876226653599977 |
| 2 PTH→RAMP2 | 0.872822134930536 |
| 2 LTB→CD40 | 0.871951342637788 |
| 2 COL9A2→GP6 | 0.868674486042229 |
| 2 MFGE8→ITGB3 | 0.854798187484758 |
| 3 THBS2→NOTCH3 | 0.982407671547954 |
| 3 HSPG2→ITGB1 | 0.953999514679573 |
| 3 COL3A1→DDR2 | 0.916969674081505 |
| 3 COL1A1→DDR2 | 0.909367701335782 |
| 3 COL18A1→KDR | 0.904557243899036 |
| 3 LAMB1→ITGA1 | 0.898943542504309 |
| 3 NID1→ITGAV | 0.896277116578857 |
| 3 MMP2→PECAM1 | 0.877768034735933 |
| 3 ADAM12→ITGB1 | 0.86683594435099 |
| 3 FN1→SDC2 | 0.866490708393639 |
| 3 COL6A3→SDC1 | 0.850023970146061 |
| 3 VCAN→ITGB1 | 0.8359503815125 |
| 3 COL4A2→CD93 | 0.834861043836592 |
| 3 COL18A1→ITGA5 | 0.828047535066019 |
| 3 AGRN→ITGB1 | 0.823523779591486 |
| 3 ITGAV→THY1 | 0.819229502434896 |
| 3 LUM→ITGB1 | 0.81619424748867 |
| 3 COL1A2→CD93 | 0.814088042938217 |
| 3 APP→LRP1 | 0.813910538276953 |
| 3 FBN1→ITGB1 | 0.812940373636083 |
| 4 CALM1→AQP1 | 1.71931574688756 |
| 4 GNAS→AVPR2 | 1.58284988499555 |
| 4 LGALS3→ENG | 1.50858715063185 |
| 4 CXCL12→ITGB1 | 1.4598593115103 |
| 4 VWF→ITGB1 | 1.43970171354322 |
| 4 APP→CD74 | 1.41711014335252 |
| 4 SPARC→ENG | 1.40752032842075 |
| 4 APP→LRP10 | 1.4014448996189 |
| 4 TIMP3→MET | 1.3970385430272 |
| 4 TIMP3→KDR | 1.39062176635385 |
| 4 ADAM15→ITGB1 | 1.36378604982644 |
| 4 MFGE8→PDGFRB | 1.34146403192861 |
| 4 CALM1→PDE1A | 1.33134822565414 |
| 4 CD48→CD2 | 1.32359176197016 |
| 4 ANXA2→ROBO4 | 1.31057258554493 |
| 4 ACTR2→ADRB2 | 1.29991490442386 |
| 4 HLA-C→CD8B | 1.2716429062981 |
| 4 HLA-A→CD8B | 1.27088015670532 |
| 4 HLA-E→CD8B | 1.25519959801409 |
| 4 LAMA5→BCAM | 1.25153888728164 |
| 5 LGALS1→ITGB1 | 1.33468648674394 |
| 5 TIMP1→CD63 | 1.25369028625109 |
| 5 LGALS1→CD69 | 1.24222563437295 |
| 5 HLA-B→CD3D | 1.1784852098724 |
| 5 HSPG2→ITGB1 | 1.10544796496223 |

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|---|---|
| 5 LGALS1→PTPRC | 1.07555277067349 |
| 5 B2M→KLRD1 | 1.03648789599315 |
| 5 FN1→SDC2 | 1.01626159479641 |
| 5 VWF→ITGB1 | 0.976573980603535 |
| 5 HLA-A→APLP2 | 0.947016653301763 |
| 5 COL4A2→CD93 | 0.935313373820616 |
| 5 VCAN→ITGB1 | 0.927800879661493 |
| 5 TIMP2→ITGB1 | 0.926093051934917 |
| 5 COL4A1→CD93 | 0.913597410956282 |
| 5 HMGB1→CXCR4 | 0.887767367133365 |
| 5 VIM→CD44 | 0.868309890874858 |
| 5 SPON2→ITGB1 | 0.859894064798373 |
| 5 SPARC→ENG | 0.857490483579659 |
| 5 APP→CD74 | 0.851189287759841 |
| 5 CALM1→INSR | 0.84096886769599 |
| 6 INSL3→RXFP2 | 0.825456047771773 |
| 6 ENHO→GPR19 | 0.818981728632694 |
| 6 ASIP→MC4R | 0.778344113790869 |
| 6 NMB→BRS3 | 0.764813172216032 |
| 6 ARF1→PLD2 | 0.761592833252543 |
| 6 LHB→ADRB3 | 0.750930480338519 |
| 6 INSL3→TSHR | 0.75057049160016 |
| 6 AVP→TSHR | 0.743263144729492 |
| 6 EFNA1→EPHA1 | 0.740971455185814 |
| 6 FGF16→FGFR4 | 0.738218512036551 |
| 6 CALCA→ADRB3 | 0.737212369198785 |
| 6 GNB3→GABBR2 | 0.736493682969774 |
| 6 WNT1→RYK | 0.729675011578767 |
| 6 ANGPTL2→LILRB3 | 0.713688264893849 |
| 6 RLN2→RXFP2 | 0.712082485616225 |
| 6 ADAM7→ITGB7 | 0.710741572460619 |
| 6 TNFSF4→TNFRSF4 | 0.708542634831125 |
| 6 NRG3→ERBB3 | 0.706608093343043 |
| 6 SHH→GPC5 | 0.705539053647593 |
| 6 ANGPTL1→LILRB3 | 0.705041219698284 |
| 7 GNAI2→C5AR1 | 0.7057400205605 |
| 7 IFNA1→CR2 | 0.702250689436757 |
| 7 RETN→CAP1 | 0.688586674886509 |
| 7 COL4A6→CD47 | 0.683844518824982 |
| 7 PSG9→DSCAM | 0.678078154899609 |
| 7 GNAI2→FPR1 | 0.672707136424034 |
| 7 HDC→HRH3 | 0.667859881567108 |
| 7 GNAI2→CHRM1 | 0.659497577732044 |
| 7 CXCL13→CCR10 | 0.654158146403055 |
| 7 CCL7→CCR10 | 0.646535593927906 |
| 7 FGF6→SDC4 | 0.639026069116347 |
| 7 LGALS9→CD47 | 0.629745161178398 |
| 7 CLEC11A→ITGA11 | 0.625774550570712 |
| 7 HLA-A→KIR2DL3 | 0.62402941640529 |
| 7 CCL16→HRH4 | 0.616807913613931 |
| 7 ICAM4→ITGA4 | 0.614877976864002 |
| 7 HDC→HRH4 | 0.614124139879813 |
| 7 FGF23→FGFR2 | 0.611278099980327 |

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|---|---|
| 7 DNAJB11→PRTG | 0.609432620286761 |
| 7 TNFSF13→TNFRSF13B | 0.604494735270735 |
| 8 CALM3→EGFR | 0.388575738497864 |
| 8 EFEMP1→EGFR | 0.379333824528529 |
| 8 COL5A1→SDC3 | 0.374169960945968 |
| 8 DCN→EGFR | 0.373637940897607 |
| 8 VCAN→EGFR | 0.371937968081631 |
| 8 APP→TNFRSF21 | 0.371365439525868 |
| 8 TNC→SDC4 | 0.367127963225382 |
| 8 TNC→ITGA5 | 0.354937218053685 |
| 8 APP→LRP1 | 0.34874052268575 |
| 8 SFRP1→FZD6 | 0.345954957110886 |
| 8 PSAP→LRP1 | 0.344881991205425 |
| 8 GNAI2→EGFR | 0.340555256646737 |
| 8 CALR→LRP1 | 0.339384703584582 |
| 8 RGMA→BMPR2 | 0.339373910255583 |
| 8 LRPAP1→LRP1 | 0.339297962420028 |
| 8 CALM3→AQP6 | 0.338230860250393 |
| 8 FGF7→FGFR4 | 0.337386944760326 |
| 8 TNC→SDC1 | 0.333174089326784 |
| 8 FGF11→FGFR1 | 0.333006204680506 |
| 8 ST6GAL1→EGFR | 0.33160068369032 |
| 9 HLA-DRB5→CD4 | 0.534955265875734 |
| 9 HLA-F→LILRB1 | 0.524606388664092 |
| 9 CD14→ITGA4 | 0.522917697150232 |
| 9 APOE→LRP1 | 0.499359780812636 |
| 9 C1QB→LRP1 | 0.473347452365672 |
| 9 CD14→ITGB2 | 0.472756494348397 |
| 9 CD14→ITGB1 | 0.458851358572909 |
| 9 PSAP→LRP1 | 0.42835704991224 |
| 9 LGALS9→LRP1 | 0.425404546472401 |
| 9 SIRPA→CD47 | 0.421735523326683 |
| 9 CD47→SIRPA | 0.421735523326683 |
| 9 HLA-DMB→CD4 | 0.414681179133794 |
| 9 CD86→CD28 | 0.410260800550859 |
| 9 CD28→CD86 | 0.410260800550859 |
| 9 SERPINA1→LRP1 | 0.40922592530983 |
| 9 HMGB1→TLR2 | 0.408322389425709 |
| 9 ADAM10→TREM2 | 0.406725386517967 |
| 9 HLA-A→LILRB1 | 0.406666405939433 |
| 9 LRPAP1→LRP1 | 0.404909659399273 |
| 9 HLA-C→LILRB1 | 0.404815212092966 |